

Geometrically Inductive Self-Supervised Learning for EEG

A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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Problem: EEG is non-stationary – and the geometry is what changes

EEG varies on **three geometric axes**:

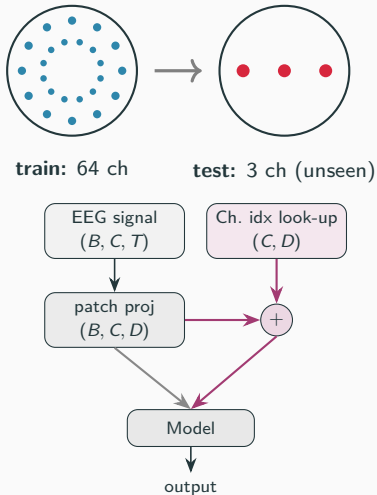
- **Session** — cap shifts day to day.
(*motivates*)
- **Subject** — head sizes differ.
(*motivates*)
- **Montage** — 2 to 256 electrodes in different layouts. ← **what I test**

Codex models (LaBraM, EEGPT):

identity-indexed embeddings — **cannot even run** on an unseen layout.

Channel-independent (BIOT): no spatial structure at all.

Neither uses scalp geometry as an *input*.



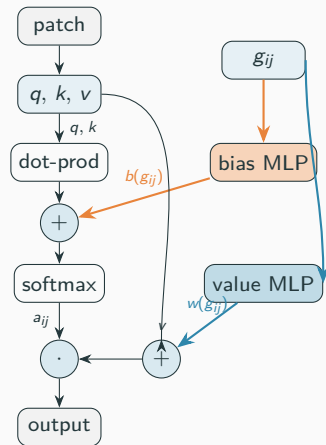
Solution: spatial attention conditioned on g_{ij}

$$g_{ij} = [p_i, p_j, p_i - p_j, \|p_i - p_j\|] \in \mathbb{R}^{10}$$

Example: $g_{C3 \rightarrow Cz}$ (normalized coords)

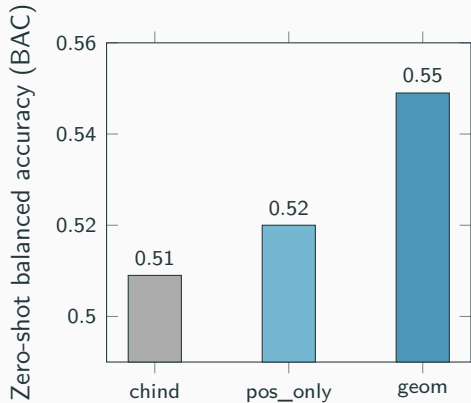
p_i (C3)	(x, y, z)	(-0.71, 0.00, 0.59)
p_j (Cz)	(x, y, z)	(0.00, 0.00, 0.95)
$p_i - p_j$	disp.	(-0.71, 0.00, -0.36)
$\ p_i - p_j\ $	dist.	0.80

$3 + 3 + 3 + 1 = 10$ dims. **pos_only** ablation keeps just $[p_i, p_j] \in \mathbb{R}^6$.



bias path only = G1, **value path** only = G2, both = G3.
Ablation: the signal is in the **value path** — $G2 \approx G3 \gg G1$ ($G1 \approx$ no geometry).

Headline: at scale, geometric beats chind on zero-shot transfer



cells = row-normalized recall (%); diag mean = BAC

geom (G2, full) (BAC 0.549)

	Pred L	Pred R
T.L	56.4	43.6
T.R	46.6	53.4

pos_only (G2) (BAC 0.520)

	Pred L	Pred R
T.L	53.2	46.8
T.R	49.3	50.7

78 PhysioNet + 78 Sleep-EDFx pretrain → zero-shot to BCIC-2B

Conclusion: a clean positive, with clear edges

What I found:

- Geometry-conditioned attention **transfers zero-shot** to an unseen montage and **beats chind significantly**.
- Ablation localizes the effect: geometry helps through the **value path** ($G2 \approx G3 \gg G1$).

Honest limits:

- Demonstrated on **one** held-out montage (BCIC); a learned-codex baseline and broader montage coverage remain open.

Future work: second distinct-montage held-out (e.g. 128 ch); larger corpus; implement the codex baseline.

Did I run what the proposal promised?

Proposed experiment	Outcome
E2(c) — cross-montage zero-shot	✓ the headline win, significant
E3 — where g_{ij} enters (G1/G2/G3)	✓ signal is value-path: $G2 \approx G3 \gg G1$
E4 — channel-independent baseline	✓ promoted to my baseline
E1 — in-distribution vs. codex	codex can't run on a new montage
E2(a) — cross-session	session drift is physiological, not geometric
E2(b) — cross-subject / head size	no public head-size-varied dataset

Top: delivered. **Bottom:** out of reach for a course project — reported as limits.

Key References

- Wang et al. **EEGPT**. NeurIPS 2024 – EEG foundation model; training recipe.
- Jiang et al. **LaBraM**. ICLR 2024 – codex (identity-indexed) baseline.
- Yang et al. **BIOT**. NeurIPS 2023 – channel-independent baseline.
- Schalk et al. 2004 – PhysioNet MI dataset.
- Tangermann et al. 2012 – BCIC-2B dataset.
- Kemp et al. 2000 – Sleep-EDFx dataset.

Full reference list and per-experiment details in the project report.